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Unifying Global and Local Features

A Modern Approach to Sports Video Analysis

Hieu Tran | Amberg

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1. Introduction

Task Overview

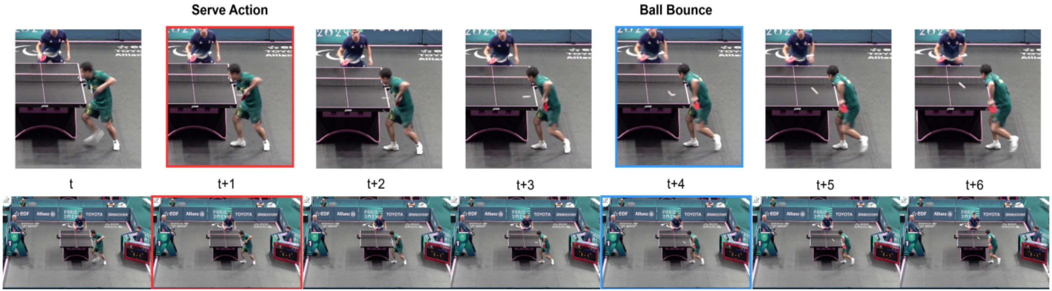

Fig. 3. Example of Precise Event Spotting: in table tennis, the moment a player contacts the ball during a serve (red) or when a ball bounces on the table (blue).

Xu H, Baniya AA, Well S, Bouadjenek MR, Dazeley R, Aryal S. Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges. arXiv:2505.03991, 2025.

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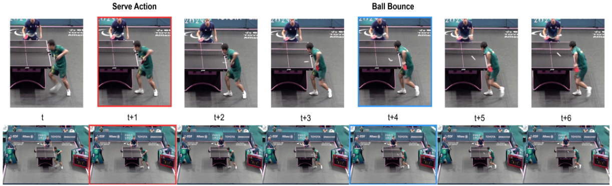

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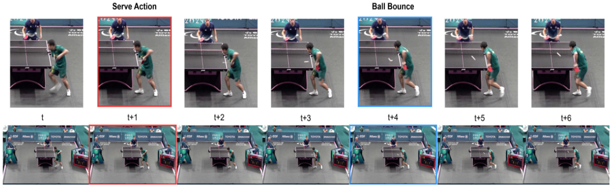
- Goal: Detect *precise timestamps* of events, not long temporal segments



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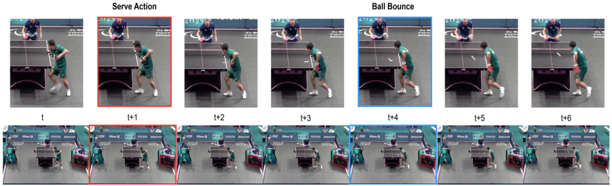
- Domain: Action spotting in sports videos (soccer, diving, gymnastics)
- Goal: Detect *precise timestamps* of events, not long temporal segments
- Applications:
  - Highlight generation
  - Video analytics and broadcasting automation
  - Downstream temporal retrieval



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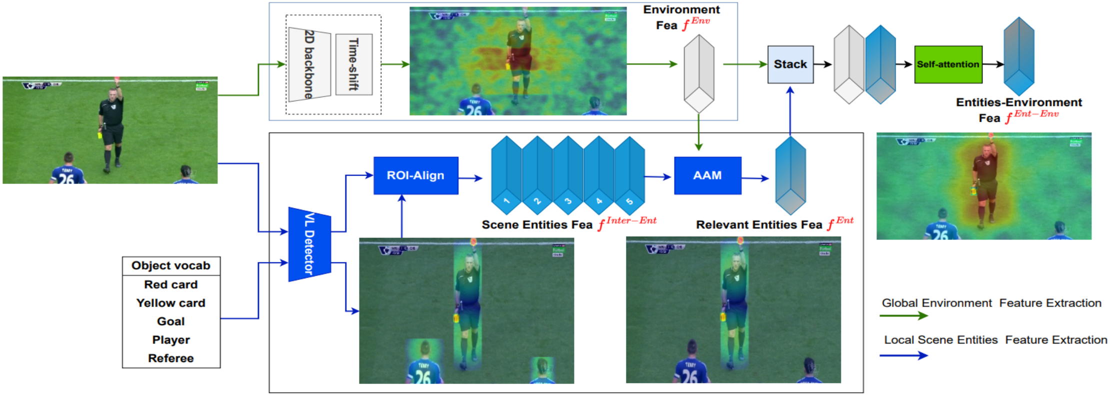
- Standard pipelines rely on global frame features only
- Current methods behave as black boxes, missing small but critical entities
- Two-phase approaches: heavy 3D backbones, multiple extractors
- End-to-end methods (e.g., E2E-Spot): lighter but still global-only

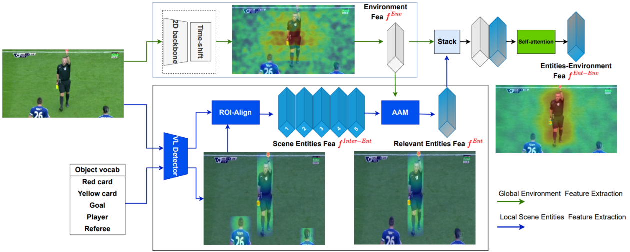
2. UGL Architecture

Global Environment Feature



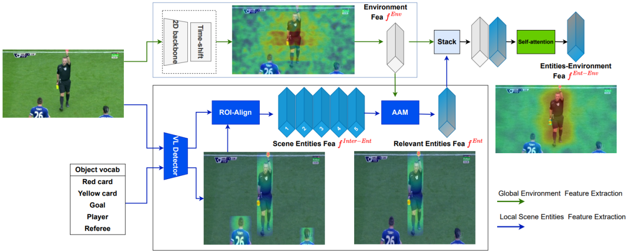
2. UGL Architecture
Global Environment Feature

- Backbone: RegNet-Y (efficient 2D CNN design)



2. UGL Architecture
Global Environment Feature

- Backbone: RegNet-Y (efficient 2D CNN design)
- Temporal Integration: Gate-Shift Module (GSM)
  - Shifts feature maps forward and backward along temporal axis
  - Shares information across frames
  - Lightweight alternative to 3D CNNs



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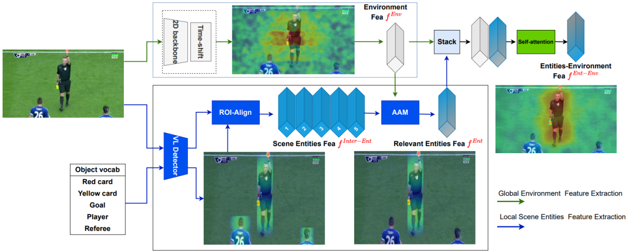
- Backbone: RegNet-Y (efficient 2D CNN design)
- Temporal Integration: Gate-Shift Module (GSM)
  - Shifts feature maps forward and backward along temporal axis
  - Shares information across frames
  - Lightweight alternative to 3D CNNs
- Output: Global environment feature vector per frame capturing spatial and short-term temporal context



2. UGL Architecture

Local Scene Entities with GLIP

Why GLIP?

Global features alone often ignore small but crucial entities (cards, ball, etc.), especially under class imbalance

- GLIP: Grounded Language-Image Pre-training (training-free detector)
- Input: Frame image + text prompt listing sports entities
  - *"ball, yellow card, red card, referee, . . . "*
- Mechanism:
  - Language encoder processes text (BERT)
  - Visual encoder produces image proposals (Swin backbone)
  - Cross-modal multi-head attention aligns proposals with textual entities

2. UGL Architecture

Adaptive Attention Mechanism (AAM)

Problem

Not all detected entities are equally relevant for the current action

AAM Solution

Takes both local entity features and global environment feature:

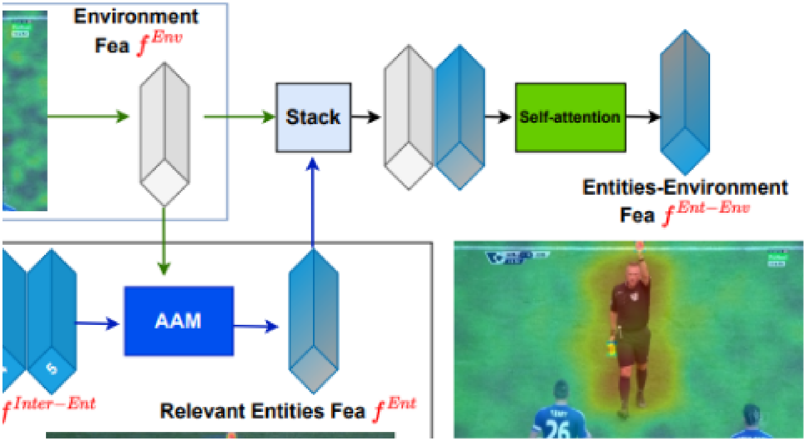
1. Hard attention: Select most promising entities
2. Soft self-attention: Fuse them into single compact representation

Output

Local relevant scene entities feature emphasizing objects crucial for the action

2. UGL Architecture

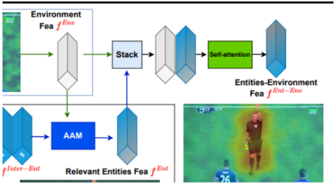
Fusion of Global and Local



2. UGL Architecture

Fusion of Global and Local

1. Concatenate:
   - Global environment feature
   - Local relevant entities feature



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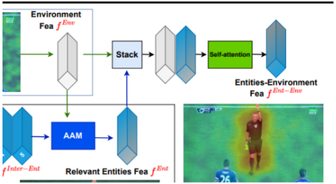
Fusion of Global and Local

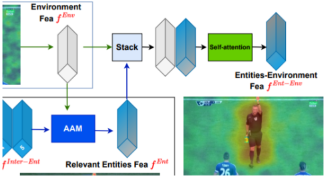
1. Concatenate:
   - Global environment feature
   - Local relevant entities feature
2. Apply:
   - Self-attention layer to model interactions
   - Average pooling for compactness



2. UGL Architecture

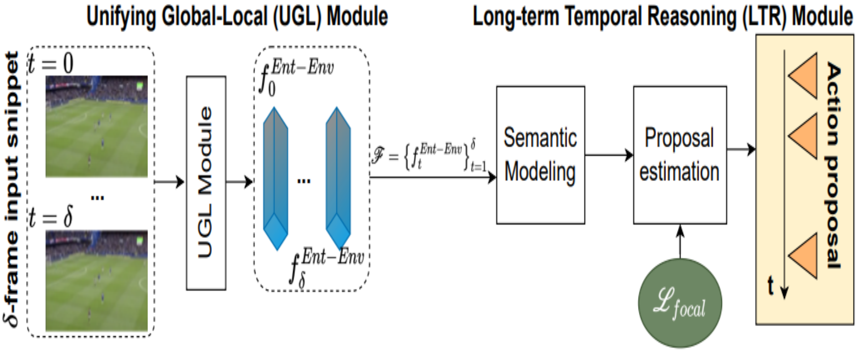
Fusion of Global and Local

1. Concatenate:
   - Global environment feature
   - Local relevant entities feature
2. Apply:
   - Self-attention layer to model interactions
   - Average pooling for compactness
3. Output: Unified entities and environment representation per frame capturing both background context and key object semantics



2. UGL Architecture

Long-Term Temporal Reasoning (LTR)

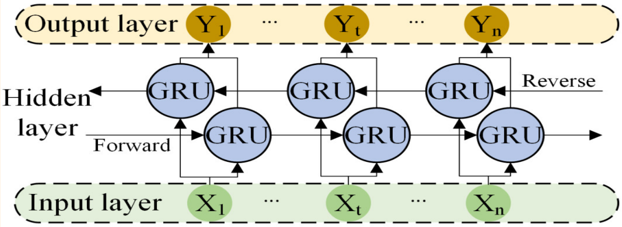


2. UGL Architecture

Long-Term Temporal Reasoning (LTR)

Temporal Modeling

- Input: Sequence of fused frame features for T-frame snippet
- Model: 1-layer **bidirectional GRU** capturing semantic relationships
- Prediction: Per-frame class scores via fully connected + softmax



Liu JH, Yang WH, He J, Wang ZM, Jia L, Zhang CF, Yang WW. Intelligent prediction of rail corrugation evolution trend based on self-attention bidirectional TCN and GRU. Intell. Robot. 2024;4(4):318–338.

3. Benchmark Datasets

Three Sports Domains

SoccerNet-v2

- $\sim$550 matches
- $\sim$764 hours of video
- $>$300k labeled spots
- Strong clutter, camera changes

FineDiving

- 3,000 diving clips
- Phase transitions (entry, twist, etc.)

FineGym

- 5,374 gym performances
- 32 spotting classes
- Converted from action segments to timestamps

4. Quantitative Results

SoccerNet-v2; FineDiving & FineGym

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Tabelle: SoccerNet-v2 challenge quantitative comparison (T-mAP).

Method	T-mAP (%)	Gap to UGL (%)	Notes
UGL (ours)	**69.38**	–	current SOTA
E2E-Spot	66.73	-2.65	previous SOTA
Soares (two-phase)	67.81	-1.57	1st 2022

Tabelle: FineDiving and FineGym improvements of UGL over E2E-Spot (no optical flow).

Dataset	Δ mAP (Gap to E2E-Spot) (%)
FineDiving	87.7 (+2.4)
FineGym	67.8 (+1.3)

5. Conclusion

- Global + Local entity features to produce precise, interpretable per-frame representations
- Top result on SoccerNet-v2: **69.38% T-mAP**
- GRU + Focal Loss handles temporal reasoning and class imbalance effectively

Thank you for listening!

Questions are welcome

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- Purpose: add short-term temporal context to the global branch without using a heavy 3D CNN
- GSM: shifts feature maps forward and backward along the temporal axis
- Output: one global environment feature vector per frame that captures spatial context and neighbor-frame information

Appendix

B. RoIAlign in GLIP

- Input: one video frame and a text prompt such as “ball, yellow card, red card, referee”
- GLIP: finds possible object locations by matching the words with what it sees in the image
- RoIAlign: turns each detected box into the same fixed-size feature map, like cutting out each object and resizing it so the network can compare them fairly
- Why it matters: the ball may be small, while a player or referee may be larger, but RoIAlign keeps the important details from each box
- Output: a set of candidate entity features that the local scene entities branch can use later

Appendix

C. Adaptive Attention Mechanism (AAM)

- Problem: not every detected entity is relevant for the current action
- Hard attention: selects the most promising entities from the GLIP proposals
- Soft self-attention: fuses the selected entities into a compact local relevant scene entities feature
- Goal: emphasize the ball, cards, referee, or other key objects that matter for spotting